Practical - 8

MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

A. Theory

Is the schema normalized properly?

Yes. The schema is properly structured into separate tables such as students, events, and registrations, which reduces duplicate data. The registrations table handles the many-to-many relationship between students and events.

Are primary keys and foreign keys correctly defined?

Yes. Each table has its own Primary Key, such as student_id, event_id, and registration_id. The registrations table uses student_id and event_id as Foreign Keys to connect the related tables.

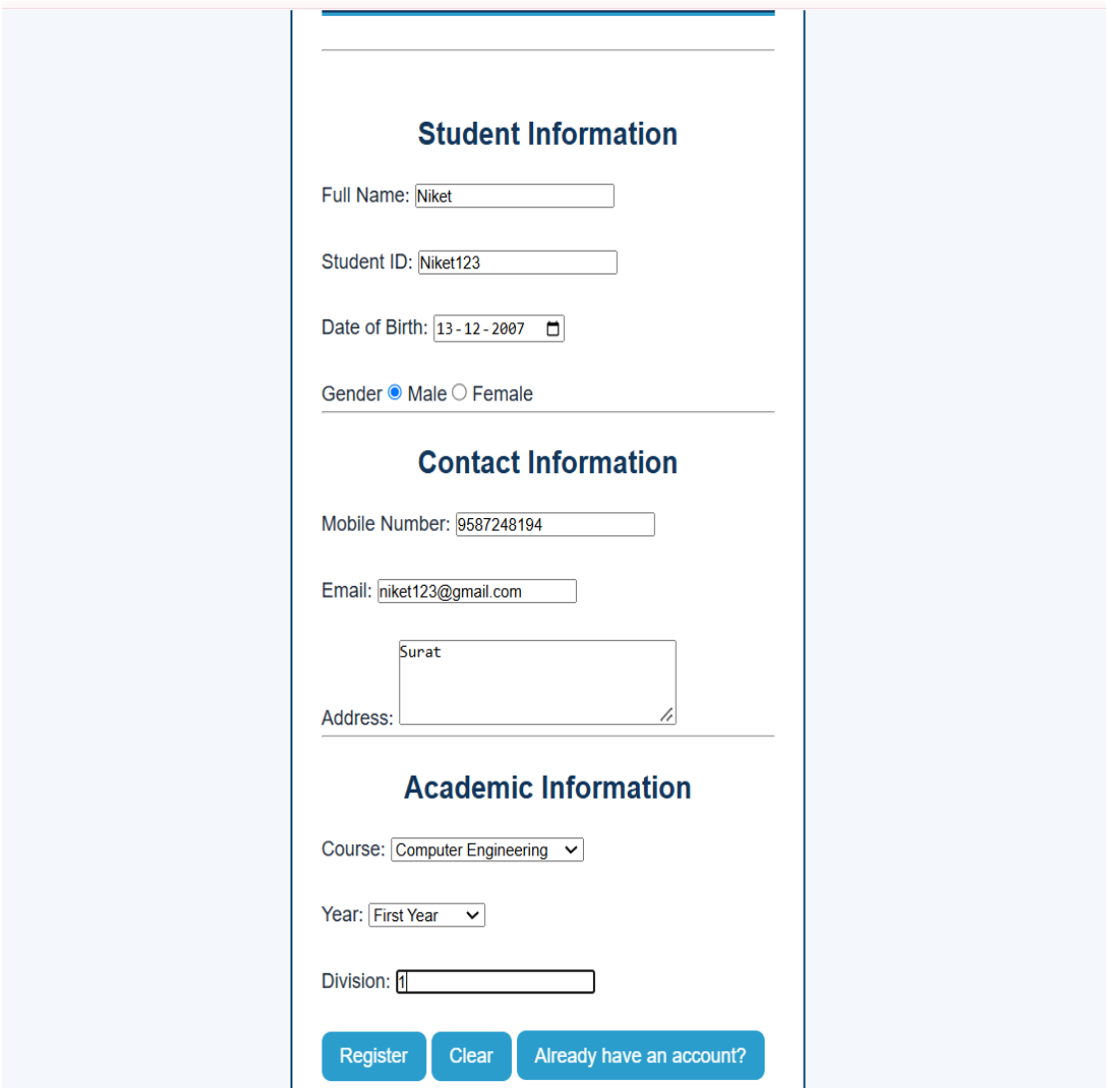
Is the database connection handled securely?

Yes. Practical 8 uses PDO, prepared statements, and prepare()/execute() instead of directly inserting user input into SQL queries. This helps prevent SQL Injection. Passwords are also stored using password_hash() rather than as plain text.

B. Implementation

Created three MySQL tables: students, events, and registrations.
Defined Primary Keys and Foreign Keys to establish relationships between the tables. Connected PHP with MySQL using PDO.
Used prepared statements with prepare() and execute() to insert and retrieve data securely.
Used password_hash() to store passwords securely instead of saving plain-text passwords.
Stored student registration and event-related data in the relational database.

C. Evidence of Work (Screenshots)



registration_id	student_id	form_student_id	full_name	date_of_birth	gender	mobile	email	address	course	academic_year	division	registered_at
1	1	Niket123	Niket	2008-12-13	Male	9587248194	niket123@gmail.com	Surat	Computer Engineering	First Year	1	2026-10-19 20:37:30

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	id	title	event_date	location
Edit Copy Delete	1	Barracks	2026-11-07	Campus Grounds
Edit Copy Delete	2	National Hackathon	2026-11-21	Innovation Centre
Edit Copy Delete	3	Tech Fest	2026-12-05	Main Auditorium
Edit Copy Delete	4	Career Guidance Workshop	2026-12-12	Seminar Hall

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Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

	id	student_id	full_name	date_of_birth	gender	mobile	email	address	course	academic_year	division
Edit Copy Delete	1	Niket123	Niket	2008-12-13	Male	9587248194	niket123@gmail.com	Surat	Computer Engineering	First Year	1
Edit Copy Delete	2	STU1001	Aarav Shah	2005-04-12	Male	9876501001	aarav.shah@example.com	Mumbai	Computer Engineering	Second Year	1
Edit Copy Delete	3	STU1002	Anaya Patel	2006-08-23	Female	9876501002	anaya.patel@example.com	Pune	Information Technology	First Year	2
Edit Copy Delete	4	STU1003	Kabir Desai	2004-11-05	Male	9876501003	kabir.desai@example.com	Nashik	Mechanical Engineering	Third Year	1
Edit Copy Delete	5	STU1004	Ira Kulkarni	2005-02-17	Female	9876501004	ira.kulkarni@example.com	Nagpur	Computer Engineering	Second Year	2

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D. GitHub Repository Link:

https://github.com/25dce037-tech/Student-Hub-Portal.git

E. Challenges Faced:

Understanding the relationship between students, events, and registrations.
Correctly defining Primary Keys and Foreign Keys.
Handling database connection errors using PDO.
Writing SQL queries with prepared statements and placeholders.
Avoiding duplicate records and maintaining data consistency.
Understanding the difference between password hashing and normal encryption.

F. Learning Outcomes:

Learned how to design a basic relational database schema.
Understood Primary Keys, Foreign Keys, and table relationships.
Learned how PHP communicates with MySQL using PDO.
Learned how prepared statements prevent SQL Injection.
Learned secure password storage using password_hash().
Gained practical knowledge of storing and managing dynamic website data in a real database instead of JSON/CSV files.

Faculty Remarks: